KAIZEN MATHS

Lesson starter

Linear Equations

Challenging Linear Equations

Name _____

S1

$4x + 18 = 2x + 8$

S2

$5(x+10) = 10(x+2) + 75$

S3

$5\big(x + (9 - 3)\big) = 3(x+2) + 32$

S4

$12x + 16 = 5x + 65$

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E1

$4(x+4) = 2(x+10) + 0$

E2

$8\big(x + (6 - 10)\big) = 3(x+8) + -46$

E3

$5x + 3 = 10x + 23$

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Teacher answers & working

Linear Equations · Challenging Linear Equations

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S1

$4x + 18 = 2x + 8$

Answer:

$x = -5$

Working:

Subtract $2x$ from both sides.

$2x + 18 = 8$

Subtract 18 from both sides.

$2x = -10$

Divide both sides by 2.

$x = -5$

S2

$5(x+10) = 10(x+2) + 75$

Answer:

$x = -9$

Working:

Distribute on both sides.

$5x + 50 = 10x + 20 + 75$

Subtract $10x$ from both sides.

$-5x + 50 = 95$

Subtract 50 from both sides.

$-5x = 45$

Divide both sides by -5.

$x = -9$

Teacher answers & working

Linear Equations · Challenging Linear Equations

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S3

$5\big(x + (9 - 3)\big) = 3(x+2) + 32$

Answer:

$x = 4$

Working:

Expand parentheses and distribute.

$5x + 30 = 3x + 6 + 32$

Subtract $3x$ from both sides.

$2x + 30 = 38$

Subtract 30 from both sides.

$2x = 8$

Divide both sides by 2.

$x = 4$

S4

$12x + 16 = 5x + 65$

Answer:

$x = 7$

Working:

Subtract $5x$ from both sides.

$7x + 16 = 65$

Subtract 16 from both sides.

$7x = 49$

Divide both sides by 7.

$x = 7$

Teacher answers & working

Linear Equations · Challenging Linear Equations

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E1

$4(x+4) = 2(x+10) + 0$

Answer:

$x = 2$

Working:

Distribute on both sides.

$4x + 16 = 2x + 20 + 0$

Subtract $2x$ from both sides.

$2x + 16 = 20$

Subtract 16 from both sides.

$2x = 4$

Divide both sides by 2.

$x = 2$

E2

$8\big(x + (6 - 10)\big) = 3(x+8) + -46$

Answer:

$x = 2$

Working:

Expand parentheses and distribute.

$8x + -32 = 3x + 24 + -46$

Subtract $3x$ from both sides.

$5x + -32 = -22$

Subtract -32 from both sides.

$5x = 10$

Divide both sides by 5.

$x = 2$

Teacher answers & working

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E3

$5x + 3 = 10x + 23$

Answer:

$x = -4$

Working:

Subtract $10x$ from both sides.

$-5x + 3 = 23$

Subtract 3 from both sides.

$-5x = 20$

Divide both sides by -5.

$x = -4$